



NSBM Green University

Faculty of Computing

CN204.3 -Network Security

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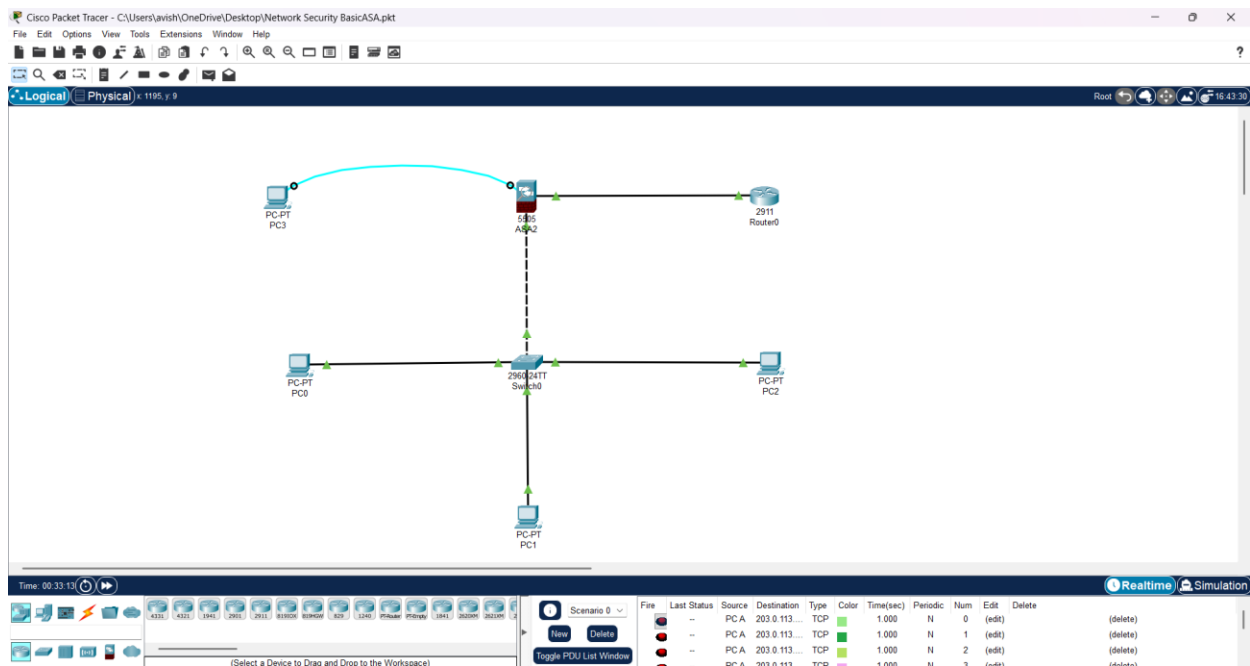
Course Work

Group R

1. Introduction

- This project focuses on the configuration of a Cisco ASA firewall to secure an internal LAN from an external router, establishing a controlled security boundary. The aim was to validate basic connectivity and then enforce access control through ACLs. Initially, ICMP traffic was tested to confirm routing and communication. Once verified, ACLs were configured to allow only HTTP traffic while blocking all other protocols

2. Network Topology



➤ The network consisted of:

- An ASA firewall configured with inside and outside VLAN interfaces.
- An internal LAN with PCs configured in the inside network.
- An external router simulating Internet access.

3. Interface and Routing Configuration

ASA Interface Verification

Commands used:

```
show interface ip brief
show switch vlan
show route
```

ASA2

Physical Config CLI Attributes

IOS

ASA1(config)#show running-config
: Saved
:
ASA Version 8.4(2)
!
hostname ASA1
enable password XK9CbN4MkQEIOjdt encrypted
ASA1>enable
Password:
ASA1#show switch vlan

VLAN Name	Status	Ports
1 inside	up	Et0/1, Et0/2, Et0/3, Et0/4 Et0/5, Et0/6, Et0/7
2 outside	up	Et0/0

ASA1#

ASA1(config)#show interface ip brief

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	unassigned	YES	unset	up	up
Ethernet0/1	unassigned	YES	unset	up	up
Ethernet0/2	unassigned	YES	unset	down	down
Ethernet0/3	unassigned	YES	unset	down	down
Ethernet0/4	unassigned	YES	unset	down	down
Ethernet0/5	unassigned	YES	unset	down	down
Ethernet0/6	unassigned	YES	unset	down	down
Ethernet0/7	unassigned	YES	unset	down	down
Vlan1	192.168.1.1	YES	manual	up	up
Vlan2	203.0.113.1	YES	manual	up	up

ASA1(config)#
ASA1(config)#show route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is 203.0.113.254 to network 0.0.0.0
C 192.168.1.0 255.255.255.0 is directly connected, inside, Vlan1
C 203.0.113.0 255.255.255.0 is directly connected, outside, Vlan2
S* 0.0.0.0/0 [1/0] via 203.0.113.254
ASA1(config)#

- Results confirmed that inside and outside interfaces were active, VLANs were properly assigned, and a static default route was configured.

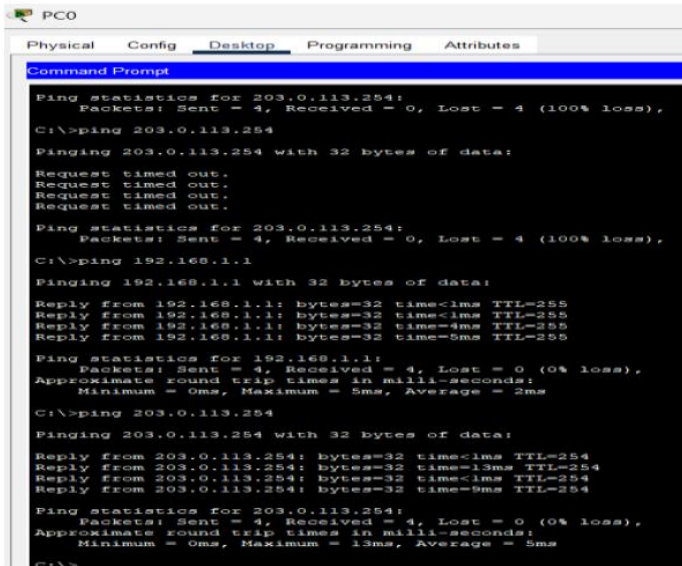
Router Configuration Verification

Commands executed:

```
show ip interface brief
show ip route
```

Output verified Router0 had IP 203.0.113.254, interfaces were up, and a static route to the internal LAN was present.

4. Baseline Connectivity (Before ACLs)



The screenshot shows a Windows Command Prompt window titled 'PC0'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is active, showing a 'Command Prompt' window. The output of the command prompt shows the results of several ping commands. First, a ping to 203.0.113.254 is shown, which fails with 100% loss. Then, a ping to 192.168.1.1 is shown, which succeeds with 0% loss. Finally, a ping to 203.0.113.254 is shown again, which succeeds with 0% loss.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Ping statistics for 203.0.113.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 203.0.113.254
Pinging 203.0.113.254 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 203.0.113.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=4ms TTL=255
Reply from 192.168.1.1: bytes=32 time=5ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 2ms
C:\>ping 203.0.113.254
Pinging 203.0.113.254 with 32 bytes of data:
Reply from 203.0.113.254: bytes=32 time<1ms TTL=254
Reply from 203.0.113.254: bytes=32 time=13ms TTL=254
Reply from 203.0.113.254: bytes=32 time<1ms TTL=254
Reply from 203.0.113.254: bytes=32 time=9ms TTL=254

Ping statistics for 203.0.113.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 5ms
C:\>
```

```
Router0#ping 203.0.113.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/11 ms

Router0#ping 192.168.1.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router0#
```

- Baseline ping tests confirmed that devices could communicate with each other before ACL restrictions were applied. This validated correct IP addressing and routing setup.

5. ACL Testing with ICMP (Temporary)

```
Router0#ping 203.0.113.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/1/7 ms

Router0#ping 192.168.1.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/5/13 ms

Router0#
```

- A temporary ACL was placed on the outside interface to permit ICMP traffic for connectivity testing:

```
access-list OUTSIDE-IN extended permit icmp host 203.0.113.254 192.168.1.0
255.255.255.0
access-group OUTSIDE-IN in interface outside
```

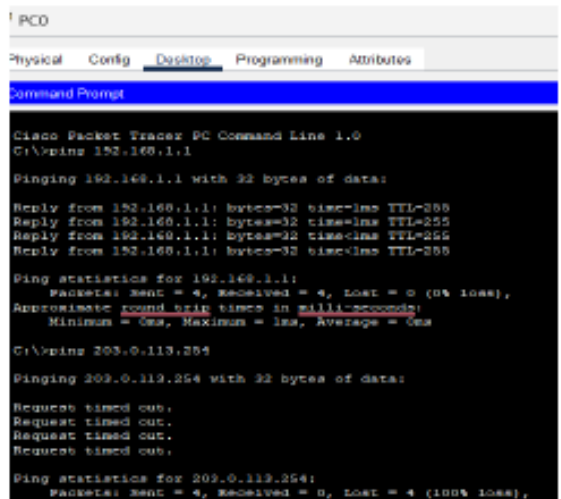
This allowed pings to verify correct connectivity before moving to stricter ACLs.

6. Final HTTP-Only ACL

- After ICMP testing, the ACL was replaced to allow only HTTP from the LAN to the outside:

```
access-list INSIDE-ONLY-HTTP extended permit tcp 192.168.1.0 255.255.255.0 any eq
www
access-list INSIDE-ONLY-HTTP extended deny ip any any
access-group INSIDE-ONLY-HTTP in interface inside
```

This ensured that HTTP traffic was permitted, while ICMP and all other protocols were blocked.



```
PCO
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 203.0.113.254

Pinging 203.0.113.254 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 203.0.113.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

ASA1(config)#show access-list
access-list cached ACL log flows: total 0, denied 0 (deny-flow-max 4096) alert-interval 300
access-list OUTSIDE-IN: 2 elements; name hash: 0xa4b08dad
access-list OUTSIDE-IN line 1 extended permit icmp host 203.0.113.254 192.168.1.0 255.255.255.0
echo(hitcnt=0) 0x55cd8704
access-list OUTSIDE-IN line 2 extended permit icmp 192.168.1.0 255.255.255.0 host 203.0.113.254
echo-reply(hitcnt=0) 0x1a1453c5
access-list INSIDE-ONLY-HTTP: 2 elements; name hash: 0x1c16544
access-list INSIDE-ONLY-HTTP line 1 extended permit tcp 192.168.1.0 255.255.255.0 any eq
www(hitcnt=36) 0x5d3c0ca5
access-list INSIDE-ONLY-HTTP line 2 extended deny ip any any(hitcnt=16) 0x4b446534
ASA1(config)#

ASA1(config)#show running-config | include access-group
access-group INSIDE-ONLY-HTTP in interface inside
ASA1(config)#
```


8. Conclusion

The project successfully demonstrated how an ASA firewall enforces access control using ACLs. Initially, ICMP ACLs validated connectivity. Later, the final ACL restricted traffic to HTTP-only, showing that the ASA can effectively filter and manage network traffic. The objectives of securing the LAN and controlling permitted protocols were fully achieved.

9. References

- Cisco Networking Academy. (2023). Introduction to Network Security. Cisco Systems.
- Odom, W. (2020). CCNA 200-301 Official Cert Guide, Volume 2. Cisco Press.
- Cisco Systems. (2022). Cisco ASA Series General Operations CLI Configuration Guide.

Retrieved from: <https://www.cisco.com>

Packet Tracer :-

https://drive.google.com/drive/folders/1Xeq0x8x9u20L0NQmViA_UJLXAPYcCp6u?usp=drive_link

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